Aura UI: Ambient Stress Sensing and Peripheral Micro-Interventions

Pilot Study Proposal, Running Prototype Verification, & Experimental Plan

Course: Human-Computer Interaction (HCI)

Institution: Institute of Business Administration (IBA)

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October 25, 2026

Abstract

Traditional digital stress management applications frequently rely on intrusive modal alerts, disrupting task immersion and amplifying cognitive friction. In this project, we introduce **Aura UI**, an ambient desktop affective computing system that passively infers real-time cognitive strain and user frustration through peripheral behavioral micro-telemetry (keystroke dynamics and mouse jitter). Rather than jarring interruptions, Aura UI employs graduated calm technology: soothing ambient display modulation (brightness and color temperature), smooth acoustic volume adjustments, an intelligent notification gatekeeper, on-demand physiological sighing breathwork, and privacy-preserving post-session LLM reflections. This document presents the verified running prototype through empirical visual UI evidence, alongside a structured experimental pilot plan to evaluate its impact on cognitive workload and task recovery.

1 System Purpose & HCI Foundation

Aura UI is built upon the paradigm of **Calm Technology** (Weiser & Brown) and **Peripheral Affective Computing**:

- **Implicit Passive Sensing:** User state is derived non-invasively without video monitoring or wearable sensors, analyzing rolling-window keystroke Coefficient of Variation ($\text{CoV} = \sigma_{\text{flight}}/\mu_{\text{flight}}$), dwell duration, error/backspace bursts, and mouse trajectory velocity/agitation.
- **Graduated Micro-Interventions:**
  - *Tier 0 (Calm, Score 0–29):* Full display luminance, cool color temperature, standard notification dispatch.
  - *Tier 1 (Mild, Score 30–59):* Subtle screen backlight dimming, smooth color temperature shift toward warm Kelvin spectra, gentle Spotify volume attenuation.
  - *Tier 2 (Elevated, Score 60–100):* Focus Assist/DND notification suppression, an interactive physiological sighing guide (double inhale, slow exhale), and proactive empathetic co-regulation.
- **Privacy by Design:** Keystroke textual content is never inspected, stored, or transmitted; only anonymous aggregate numeric statistical indicators are processed.

2 Running Application: Live Demonstration & UI Evidence

The application is fully operational as a native desktop client (runnable via `python run.py` or standalone `AuraUI.exe`). The following figures demonstrate the running application across its complete operational cycle:

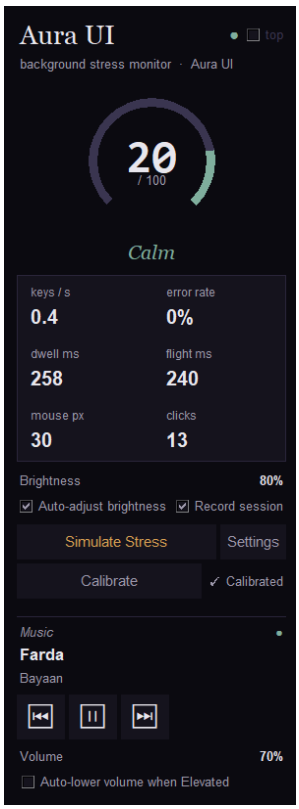


Figure 1: **Running Aura UI Desktop Dashboard**: Dual-column ambient computing interface featuring the live circular stress dial, raw micro-telemetry metrics (typing speed, error rate, dwell, flight latency, mouse pixels), auto-brightness controls, Focus notification status, Spotify integration, and conversational Aura co-regulation.



Figure 2: **Tier 1 (Mild Stress, 48/100)**: Ambient dimming active and color temperature shifted to 5000K.

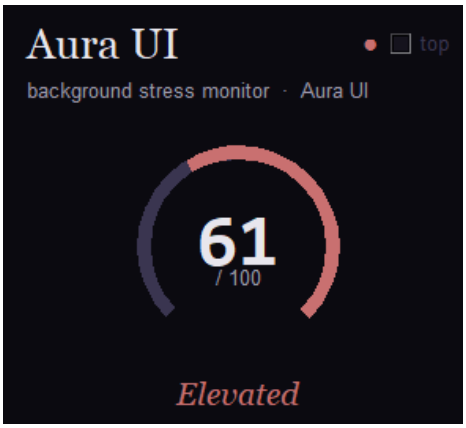


Figure 3: **Tier 2 (Elevated Stress, 85/100)**: Focus mode blocking incoming alerts; screen warmed to 3500K.



Figure 4: **Physiological Sighing Intervention**: Animated breath guide triggered during sustained elevation.

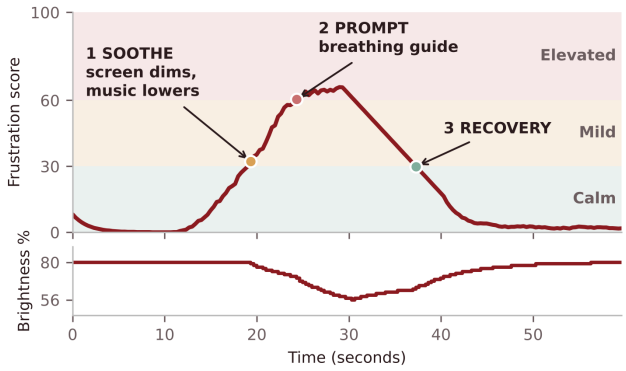


Figure 5: **Session Telemetry Plot**: Real recorded session demonstrating Soothe, Prompt, and Recovery phases.

3 Experimental Plan (Pilot Study)

3.1 Research Question & Hypotheses

- **Research Question:** Does peripheral sensory attenuation coupled with an intelligent notification gatekeeper mitigate subjective frustration and attention fragmentation during high-cognitive-load desktop computing?
- **H1 (Workload Reduction):** Participants working with active Aura UI interventions will report significantly lower perceived frustration on the NASA-TLX subscale relative to a baseline control.
- **H2 (Attention Continuity):** Notification suppression during elevated stress episodes will significantly reduce task-switching latency and typing recovery time.
- **H3 (User Acceptance):** Ambient sensory adjustments (screen dimming/warmth) will be perceived as significantly less disruptive than standard intrusive dialog notifications.

3.2 Experimental Design & Participants

- **Design:** Within-subjects (A/B) counterbalanced study design to eliminate order and learning effects.
- **Condition A (Control / Baseline):** Aura UI operates in passive logging mode (sensors record telemetry; no ambient screen shifts, no notification gating, no breath prompts).
- **Condition B (Intervention / Aura Active):** Full ambient modulation, notification suppression during high stress, and physiological sighing availability.
- **Sample Size:** $N = 8$ to 12 university students/knowledge workers undergoing computer-centric analytical problem solving.

3.3 Experimental Tasks (Standardized Cognitive Stressors)

Participants perform two equivalent 12-minute cognitively demanding knowledge tasks (counterbalanced across conditions):

- **Task Protocol:** Timed code debugging / textual synthesis with accuracy goals under temporal pressure.

- **Distraction Injector:** Aura UI's built-in notification engine triggers controlled synthetic communication pings (e.g., Slack, Email, Calendar alerts) at fixed pseudo-random intervals.

3.4 Evaluation Metrics & Instruments

Metric Category	Data Source / Instrument	Measurement Target
Objective Telemetry	Aura CSV Logger (`session_log_*.csv`)	Typing CoV, error bursts, mouse jitter, total duration in Elevated tier, time-to-recovery (seconds back to Calm).
Notification Dynamics	Session Statistics / Focus Queue	Volume of notifications suppressed vs. delivered; post-stress digest engagement.
Subjective Workload	NASA-TLX (Task Load Index)	Mental demand, temporal demand, effort, and frustration ratings (0–100 scale).
System Usability	System Usability Scale (SUS)	Overall usability, satisfaction, and perceived intrusiveness of peripheral cues.
Qualitative Insights	Post-Study Semi-Structured Interview	User sentiment on physiological sighing and LLM reflection usefulness.

Table 1: Overview of Data Collection Metrics and Instruments.

3.5 Step-by-Step Pilot Procedure

1. **Briefing & Calibration (3 mins):** Transparent briefing confirming privacy preservation (keystroke content is neither observed nor stored). The user executes Aura's 30-second typing baseline calibration.
2. **Task Session 1 (12 mins):** Participant conducts Task 1 under assigned condition (A or B) with active background telemetry logging.
3. **Mid-Study Assessment (3 mins):** Participant completes NASA-TLX for Session 1.
4. **Task Session 2 (12 mins):** Participant conducts Task 2 under the alternate condition (counterbalanced) with active telemetry logging.
5. **Post-Study Assessment & Reflection (5 mins):** Participant completes NASA-TLX for Session 2, SUS questionnaire, and inspects the Aura Journal reflection.
6. **Comparative Analysis:** Telemetry logs are rendered using `plot_session.py` to chart stress and sensory adaptation curves across both conditions.